Qualification Pack

Application Architect - Web & Mobile

Web Application Architecture/ Mobile Application Architecture

QP Code: SSC/Q8402

Version: 3.0

NSQF Level: 6

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Contents

SSC/Q8402: Application Architect - Web & Mobile

Brief Job Description

An Application Architect Web & Mobile is responsible for designing, maintaining, evaluating, and testing the architectures of web-based and mobile-based products and solutions. The individual develops enterprise-wide policies and guidelines for software implementation, governance, and compliance. Additionally, the person evaluates and selects third-party tools, APIs, and service providers to enhance and optimize software solutions.

Personal Attributes

Individuals in this role must manage and collaborate with various stakeholders while developing solution architectures for web-based or mobile-based platforms. They must have a strong analytical mindset and ensure that their technical knowledge is constantly updated.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. SSC/N8407: Define best practices, checklist, reference implementations for application design and development
2. SSC/N8306: Translate business-level use cases into technical requirements and specifications
3. SSC/N8410: Define parameters and tools to monitor and manage application performance
4. SSC/N8411: Ensure compliance of regulatory standards, best practices and policies in software development across the organization
5. SSC/N8452: Optimize Revenue, Security, and Accessibility in Web and Mobile Products
6. DGT/VSQ/N0102: Employability Skills (60 Hours)

Electives*(mandatory to select at least one):*

Elective 1: Web Application Architecture

1. SSC/N8408: Define the architecture for the web-based solution
2. SSC/N8412: Define security architecture and security controls to ensure security of the web-based solution

Elective 2: Mobile Application Architecture

1. SSC/N8409: Define the architecture for the mobile-based solution
2. SSC/N8413: Define security architecture and security controls to ensure security of the mobile application/solution

Qualification Pack (QP) Parameters

Sector	IT-ITeS
Sub-Sector	Future Skills
Occupation	Web & Mobile Development
Country	India
NSQF Level	6
Credits	20
Aligned to NCO/ISCO/ISIC Code	NCO-2015/NIL
Minimum Educational Qualification & Experience	Completed 4 year UG program (UG or diploma with courses related to Engg. / Science) OR Completed 3 year UG degree (UG or diploma with courses related to Engg. / Science) with 1.5 years of experience in relevant field.Relevant Experience (including work, internship and apprenticeship) related to Web and Mobile OR Previous relevant Qualification of NSQF Level (5) with 3 Years of experience in relevant filed.Relevant Experience (including work, internship and apprenticeship) related to Web and Mobile
Minimum Level of Education for Training in School	Not Applicable
Pre-Requisite License or Training	NIL
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	18/02/2028
NSQC Approval Date	18/02/2025
Version	3.0

Reference code on NQR	QG-06-IT-03644-2025-V2-NASSCOM
NQR Version	3.0

SSC/N8407: Define best practices, checklist, reference implementations for application design and development

Description

This unit is about developing standards and best practices for the development of solutions

Scope

The scope covers the following :

- Define standards and best practices

Elements and Performance Criteria

Standards, best practices, and design documentation

To be competent, the user/individual on the job must be able to:

PC1. conduct a comprehensive analysis of the business requirements of the organization, ensuring alignment with both functional and technical goals.

PC2. develop and implement programming and design standards for developing, maintaining, and scaling software solutions, ensuring best practices are followed.

PC3. select appropriate software tools, frameworks, and computer languages that align with organizational objectives, documenting the choices for transparency.

PC4. define standards for the allocation, management, and optimization of IT resources, ensuring alignment with business requirements.

PC5. delineate and document policies for user access management across applications, servers, and databases, ensuring adherence to security protocols.

PC6. establish enterprise processes and performance benchmarks to ensure the satisfactory performance of software products and solutions

PC7. develop and disseminate guidelines for contributing to open-source projects, if relevant to the organization, ensuring compliance with contribution standards

PC8. identify and ensure compliance with regulatory standards such as PCI DSS, HIPAA, and GDPR, among others, that the organization must adhere to

PC9. develop a cohesive operational checklist to ensure bug-free deployment and release of software, with measurable quality benchmarks

PC10. establish robust standards for source code management, control, and versioning, ensuring that code repositories are secure and scalable.

PC11. elucidate and document guidelines for maintaining a repository of successful software implementations for future reference and reuse.

PC12. specify standards and protocols to reference prior implementations when working on new projects to promote consistency and reduce errors.

PC13. define and implement a quality assurance checklist and standards to ensure compliance with software testing and review practices

PC14. select and enforce appropriate security standards and guidelines such as OWASP or other relevant security frameworks for application development.

PC15. provide a functional overview of design components to the development team, ensuring all stakeholders have a clear understanding of the architectural design and its integration into the system

PC16. create a cohesive project schedule, including weekly meetings with team members to review progress, address business requirements, and update relevant documentation, ensuring clear and measurable tracking of project deliverables.

PC17. integrate agile software development practices and DevOps principles into the application lifecycle, including iterative development, continuous integration, and automated deployment pipelines.

PC18. apply secure coding practices to ensure the integrity of the application, protect sensitive data, and prevent vulnerabilities throughout the software lifecycle

PC19. manage the entire application lifecycle, ensuring that all phases are documented and executed according to industry standards

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. the procedures and guidelines related to defining standards, policies, and best practices in software development

KU2. procedures for sharing data to ensure adherence to privacy and security regulations

KU3. how to involve appropriate stakeholders in evaluating and defining standards and regulations, ensuring a collaborative and inclusive approach

KU4. the range of standard templates, tools, and frameworks available for developing and maintaining software, and how to use them effectively

KU5. the best practices for designing, developing, and maintaining scalable, secure, and robust software solutions

KU6. different programming and design standards used to develop solutions, including how these can be adapted for specific projects

KU7. the standards and protocols for managing and controlling source code repositories, including version control and access management

KU8. the relevant compliance and regulatory standards, such as PCI DSS, HIPAA, GDPR, and others, and how to integrate them into the development process

KU9. the documentation practices for creating a Functional Overview of Design Components, ensuring team members have clear, consistent, and detailed design references

KU10. the importance of establishing a Cohesive Project Schedule with regular updates, team meetings, and documentation to address evolving business requirements

KU11. the key principles of Agile Software Development and DevOps, including iterative development, continuous integration, continuous deployment, and collaboration across teams

KU12. secure coding Practices, including methodologies for ensuring code security, preventing vulnerabilities, and adhering to security protocols like OWASP

KU13. the strategies for managing the entire application lifecycle, from design, development, and testing through to deployment, maintenance, and sunsetting

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. make informed architectural decisions that align with business goals and organizational needs

GS2. adhere to instructions, guidelines, procedures, rules, and service level agreements (SLAs)

GS3. contribute effectively to team collaboration and the overall quality of teamwork

GS4. work efficiently both independently and in collaboration with others

GS5. produce well-written, accurate work with meticulous attention to detail

GS6. evaluate business impacts, sharing relevant information with key stakeholders

GS7. communicate pertinent information clearly and promptly to others

GS8. ensure that all work is thoroughly checked and free of errors

GS9. exercise balanced judgment when faced with various situations

GS10. provide constructive feedback and detailed insights into work processes

GS11. make informed decisions on appropriate courses of action

GS12. plan and organize tasks to meet targets and deadlines effectively

GS13. utilize problem-solving strategies to address diverse challenges in different contexts

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Standards, best practices, and design documentation	**30**	**50**	**-**	**20**
PC1. conduct a comprehensive analysis of the business requirements of the organization, ensuring alignment with both functional and technical goals.	1	2	-	1
PC2. develop and implement programming and design standards for developing, maintaining, and scaling software solutions, ensuring best practices are followed.	2	3	-	1
PC3. select appropriate software tools, frameworks, and computer languages that align with organizational objectives, documenting the choices for transparency.	1	3	-	2
PC4. define standards for the allocation, management, and optimization of IT resources, ensuring alignment with business requirements.	1	2	-	1
PC5. delineate and document policies for user access management across applications, servers, and databases, ensuring adherence to security protocols.	2	3	-	1
PC6. establish enterprise processes and performance benchmarks to ensure the satisfactory performance of software products and solutions	1	3	-	1
PC7. develop and disseminate guidelines for contributing to open-source projects, if relevant to the organization, ensuring compliance with contribution standards	2	3	-	1
PC8. identify and ensure compliance with regulatory standards such as PCI DSS, HIPAA, and GDPR, among others, that the organization must adhere to	1	2	-	1
PC9. develop a cohesive operational checklist to ensure bug-free deployment and release of software, with measurable quality benchmarks	2	3	-	1
PC10. establish robust standards for source code management, control, and versioning, ensuring that code repositories are secure and scalable.	2	2	-	1

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. elucidate and document guidelines for maintaining a repository of successful software implementations for future reference and reuse.	2	3	-	1
PC12. specify standards and protocols to reference prior implementations when working on new projects to promote consistency and reduce errors.	1	3	-	1
PC13. define and implement a quality assurance checklist and standards to ensure compliance with software testing and review practices	2	3	-	1
PC14. select and enforce appropriate security standards and guidelines such as OWASP or other relevant security frameworks for application development.	2	3	-	1
PC15. provide a functional overview of design components to the development team, ensuring all stakeholders have a clear understanding of the architectural design and its integration into the system	1	2	-	1
PC16. create a cohesive project schedule, including weekly meetings with team members to review progress, address business requirements, and update relevant documentation, ensuring clear and measurable tracking of project deliverables.	2	2	-	1
PC17. integrate agile software development practices and DevOps principles into the application lifecycle, including iterative development, continuous integration, and automated deployment pipelines.	2	3	-	1
PC18. apply secure coding practices to ensure the integrity of the application, protect sensitive data, and prevent vulnerabilities throughout the software lifecycle	2	3	-	1
PC19. manage the entire application lifecycle, ensuring that all phases are documented and executed according to industry standards	1	2	-	1
NOS Total	**30**	**50**	**-**	**20**

National Occupational Standards (NOS) Parameters

NOS Code	SSC/N8407
NOS Name	Define best practices, checklist, reference implementations for application design and development
Sector	IT-ITeS
Sub-Sector	Future Skills
Occupation	Web & Mobile Development
NSQF Level	6
Credits	2
Version	2.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

SSC/N8306: Translate business-level use cases into technical requirements and specifications

Description

This unit is about gathering business requirements and translating them into technical requirements and specifications for relevant stakeholders.

Scope

The scope covers the following :

- Map requirements

Elements and Performance Criteria

Map and translate requirements

To be competent, the user/individual on the job must be able to:

PC1. define the functional and non-functional requirements of the customer, internal departments, or organization, ensuring that all needs are clearly articulated and documented

PC2. document client requirements using standardized templates, ensuring completeness, clarity, and adherence to organizational guidelines

PC3. translate business requirements into clear and actionable technical and IT specifications that align with both business goals and technological constraints

PC4. validate requirements with relevant stakeholders, both internal and external to the organization, ensuring all parties understand and agree on the proposed solution

PC5. perform business process modeling and mapping to visualize, analyze, and improve the business processes that are relevant to the defined requirements

PC6. select the appropriate technology stack to meet the defined technical and business requirements, ensuring compatibility, scalability, and alignment with the organization's long-term technology goals.

PC7. conduct a thorough risk assessment for proposed technical solutions, including the evaluation of security, scalability, and regulatory compliance risks

PC8. ensure proper mitigating measures are in place.

PC9. explore the potential of Generative AI (Gen AI) in addressing business needs and technical requirements, considering system integration and its impact on architectural decisions

PC10. apply prompt engineering techniques to effectively utilize Gen AI for system integration, business process automation, and other technical use cases, ensuring the AI is optimized for specific business challenges.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. the procedures and guidelines that relate to defining functional and non-functional requirements, ensuring compliance with internal frameworks

KU2. the policies and procedures for sharing data, including data privacy, security, and regulatory requirements
KU3. the procedures for documenting business requirements in a structured and comprehensive manner
KU4. the key technical and non-technical stakeholders, who should be involved while validating technical requirements and specifications to ensure alignment with business goals
KU5. the efficient use of standard templates and tools for documenting and managing business and technical requirements
KU6. the techniques for defining functional and non-functional requirements, including methodologies for capturing and refining customer and business needs
KU7. the best practices for documenting business requirements, ensuring clarity, completeness, and alignment with organizational standards
KU8. the approaches for defining technical requirements, including the translation of business needs into system specifications and technical designs
KU9. the principles of Business Process Modeling and Mapping, including visualization and analysis of workflows to optimize business processes
KU10. the criteria for selecting the appropriate technology stack based on business requirements, including considerations for scalability, security, and regulatory compliance
KU11. the key elements of risk assessment for new technical solutions, including the identification of security, scalability, and compliance risks, as well as methods for risk mitigation
KU12. Generative AI (Gen AI), including its capabilities, benefits, and limitations in the context of enterprise applications and system architecture
KU13. the use cases of Gen AI in application architecture, such as system integration, automation of business processes, and making architectural decisions based on AI-driven insights
KU14. the prompt engineering techniques to tailor Gen AI models to specific business requirements

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. seek clarification and advice from the appropriate stakeholders
GS2. listen actively and communicate information clearly and accurately
GS3. analyze the business impact and share relevant information with others effectively
GS4. ensure that work is thorough, accurate, and free from errors
GS5. establish and maintain positive, productive relationships with customers
GS6. perform effectively in a customer-facing environment

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Map and translate requirements	**30**	**50**	**-**	**20**
PC1. define the functional and non-functional requirements of the customer, internal departments, or organization, ensuring that all needs are clearly articulated and documented	2	3	-	1
PC2. document client requirements using standardized templates, ensuring completeness, clarity, and adherence to organizational guidelines	3	3	-	2
PC3. translate business requirements into clear and actionable technical and IT specifications that align with both business goals and technological constraints	4	4	-	1
PC4. validate requirements with relevant stakeholders, both internal and external to the organization, ensuring all parties understand and agree on the proposed solution	2	6	-	2
PC5. perform business process modeling and mapping to visualize, analyze, and improve the business processes that are relevant to the defined requirements	3	6	-	2
PC6. select the appropriate technology stack to meet the defined technical and business requirements, ensuring compatibility, scalability, and alignment with the organization’s long- term technology goals.	4	6	-	3
PC7. conduct a thorough risk assessment for proposed technical solutions, including the evaluation of security, scalability, and regulatory compliance risks	3	6	-	2
PC8. ensure proper mitigating measures are in place.	3	5	-	2
PC9. explore the potential of Generative AI (Gen AI) in addressing business needs and technical requirements, considering system integration and its impact on architectural decisions	4	6	-	3

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. apply prompt engineering techniques to effectively utilize Gen AI for system integration, business process automation, and other technical use cases, ensuring the AI is optimized for specific business challenges.	2	5	-	2
NOS Total	**30**	**50**	**-**	**20**

National Occupational Standards (NOS) Parameters

NOS Code	SSC/N8306
NOS Name	Translate business-level use cases into technical requirements and specifications
Sector	IT-ITeS
Sub-Sector	Future Skills
Occupation	Cloud Computing
NSQF Level	6
Credits	2
Version	3.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

SSC/N8410: Define parameters and tools to monitor and manage application performance

Description

This unit is about defining analyzing performance patterns as well as monitoring and reporting resource usage.

Scope

The scope covers the following :

- Analysing performance pattern
- Monitoring and reporting

Elements and Performance Criteria

Analysing performance pattern

To be competent, the user/individual on the job must be able to:

PC1. analyze the functional and non-functional requirements defined for the software solution, ensuring they are clearly understood and aligned with business goals

PC2. assess the utilization patterns of the product or services and their corresponding IT resources, identifying areas for optimization and improvement

PC3. implement methods to monitor security vulnerabilities in applications, including periodic security audits and assessments to identify potential risks

Monitor and report

To be competent, the user/individual on the job must be able to:

PC4. establish measurable Key Performance Indicators (KPIs) for monitoring the performance of the product or services, ensuring alignment with organizational objectives

PC5. outline methods for continuously monitoring application and infrastructure resource usage, focusing on performance, availability, and security

PC6. specify standards and templates for reporting usage of applications and resources, ensuring that reports are clear, actionable, and aligned with organizational goals

PC7. identify and implement appropriate tools to monitor the performance of applications, services, and IT resources, ensuring scalability and ease of use

PC8. implement tools for real-time security monitoring, such as Intrusion Detection Systems (IDS), to ensure proactive identification and mitigation of security threats

PC9. integrate performance monitoring into CI/CD pipelines, ensuring quality and security are maintained throughout the software development lifecycle

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. the procedures for analyzing functional and non-functional requirements for software solutions, ensuring alignment with business needs
KU2. the procedures for defining and implementing KPIs for monitoring the performance of software solutions
KU3. the procedures for monitoring application and infrastructure resource usage, including tools and techniques
KU4. the key stakeholders to be involved while defining KPIs and performance metrics for the solution
KU5. the effective use of standard templates and tools for monitoring performance
KU6. the best practices for monitoring and managing application performance, including methods for identifying bottlenecks and optimizing resource usage
KU7. how to assess functional and non-functional requirements to ensure the solution meets the needs of the business
KU8. how to evaluate utilization patterns of products or services, including trends in resource consumption and performance
KU9. the standards and templates for reporting application and resource usage to ensure clarity and accuracy
KU10. the methods to monitor and mitigate security vulnerabilities in applications, including vulnerability scanning, penetration testing, and regular security assessments
KU11. the tools and techniques for implementing real-time security monitoring, such as Intrusion Detection Systems (IDS), to detect and respond to potential threats quickly
KU12. how to integrate performance monitoring into CI/CD pipelines for continuous testing and validation of performance, security, and quality throughout the software development lifecycle

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Follow instructions, guidelines, procedures, rules and service level agreements
GS2. contribute effectively to team collaboration and quality output
GS3. work both independently and collaboratively, as required
GS4. produce accurate, well-written work with strong attention to detail
GS5. understand the business impact and share relevant information with others
GS6. pass on important and relevant information to the appropriate individuals
GS7. ensure that work is complete and free from errors
GS8. apply balanced judgment when assessing different situations
GS9. provide detailed and constructive feedback on work
GS10. make decisions on suitable courses of action based on analysis
GS11. plan and organize work efficiently to meet targets and deadlines
GS12. apply problem-solving strategies to handle different situations effectively

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Analysing performance pattern	**10**	**15**	**-**	**5**
PC1. analyze the functional and non-functional requirements defined for the software solution, ensuring they are clearly understood and aligned with business goals	2	3	-	1
PC2. assess the utilization patterns of the product or services and their corresponding IT resources, identifying areas for optimization and improvement	4	6	-	2
PC3. implement methods to monitor security vulnerabilities in applications, including periodic security audits and assessments to identify potential risks	4	6	-	2
Monitor and report	**20**	**35**	**-**	**15**
PC4. establish measurable Key Performance Indicators (KPIs) for monitoring the performance of the product or services, ensuring alignment with organizational objectives	2	4	-	2
PC5. outline methods for continuously monitoring application and infrastructure resource usage, focusing on performance, availability, and security	2	5	-	3
PC6. specify standards and templates for reporting usage of applications and resources, ensuring that reports are clear, actionable, and aligned with organizational goals	4	8	-	3
PC7. identify and implement appropriate tools to monitor the performance of applications, services, and IT resources, ensuring scalability and ease of use	4	8	-	2
PC8. implement tools for real-time security monitoring, such as Intrusion Detection Systems (IDS), to ensure proactive identification and mitigation of security threats	4	5	-	3

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC9. integrate performance monitoring into CI/CD pipelines, ensuring quality and security are maintained throughout the software development lifecycle	4	5	-	2
NOS Total	**30**	**50**	**-**	**20**

National Occupational Standards (NOS) Parameters

NOS Code	SSC/N8410
NOS Name	Define parameters and tools to monitor and manage application performance
Sector	IT-ITeS
Sub-Sector	Future Skills
Occupation	Web & Mobile Development
NSQF Level	6
Credits	2
Version	2.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

SSC/N8411: Ensure compliance of regulatory standards, best practices and policies in software development across the organization

Description

This unit is about ensuring legal, regulatory and compliance standards and then documenting them appropriately.

Scope

The scope covers the following :

- Legal/ Regulatory/ Compliance standards
- Documentation

Elements and Performance Criteria

Legal/ Regulatory/ Compliance standards

To be competent, the user/individual on the job must be able to:

PC1. conduct an analysis of the organization’s business and the market in which it operates

PC2. identify regulatory standards such as PCI DSS, HIPAA, and compliance requirements like GDPR that the organization must adhere to.

PC3. plan and execute audits, activities, and exercises to ensure compliance with governmental regulations, ordinances, and standards.

PC4. collaborate with cross-functional teams to regularly audit IT assets for compliance

PC5. identify and mitigate compliance risks in software development

PC6. present compliance issues to relevant stakeholders, prioritize issues, and support risk mitigation plans

PC7. monitor data sovereignty and cross-border data regulations to ensure compliance

PC8. handle and document incidents related to non-compliance, establishing appropriate corrective actions

Documentation

To be competent, the user/individual on the job must be able to:

PC9. define policies and guidelines on best IT practices that comply with necessary regulatory requirements

PC10. document and implement appropriate regulatory and compliance standards that meet the organization’s compliance needs

PC11. . create documentation on data sovereignty and cross-border regulations affecting the organization

PC12. develop and maintain a process for tracking compliance issues and mitigation efforts

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. the regulatory standards for software and IT solutions
KU2. the procedures for shortlisting third-party service providers in compliance with the applicable standards
KU3. the organizational policies for documenting various compliance standards, regulations, and guidelines
KU4. the key stakeholders involved in maintaining and reviewing system architecture for compliance
KU5. how to use standard templates and tools for documenting compliance efforts and risk assessments
KU6. the best practices for designing, developing, and maintaining software solutions while ensuring compliance
KU7. different regulatory and compliance standards like PCI DSS, HIPAA, and GDPR
KU8. how to implement necessary regulatory compliance measures effectively within the organization
KU9. the principles of data sovereignty and cross-border data regulations, and how they affect organizational data management
KU10. how to conduct business analysis and market research to inform compliance activities
KU11. how to identify and mitigate compliance risks in software development, including security, scalability, and data protection risks
KU12. how to handle incidents related to non-compliance, including escalation, reporting, and corrective actions

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. adhere to instructions, guidelines, procedures, rules, and service level agreements
GS2. contribute positively to the quality of teamwork and collaboration
GS3. effectively work both independently and as part of a team
GS4. produce accurate, well-written work with strong attention to detail
GS5. understand business impacts and communicate relevant information to stakeholders
GS6. relay pertinent information clearly and efficiently to others
GS7. ensure that all work is thoroughly reviewed and free from errors
GS8. apply balanced judgment when evaluating different situations
GS9. provide detailed and constructive feedback on work
GS10. make informed decisions on appropriate courses of action
GS11. plan and organize work effectively to meet targets and deadlines
GS12. employ problem-solving strategies to navigate various challenges

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Legal/ Regulatory/ Compliance standards	**25**	**40**	**-**	**15**
PC1. conduct an analysis of the organization’s business and the market in which it operates	2	4	-	2
PC2. identify regulatory standards such as PCI DSS, HIPAA, and compliance requirements like GDPR that the organization must adhere to.	4	5	-	2
PC3. plan and execute audits, activities, and exercises to ensure compliance with governmental regulations, ordinances, and standards.	4	6	-	2
PC4. collaborate with cross-functional teams to regularly audit IT assets for compliance	4	5	-	2
PC5. identify and mitigate compliance risks in software development	3	6	-	2
PC6. present compliance issues to relevant stakeholders, prioritize issues, and support risk mitigation plans	3	5	-	2
PC7. monitor data sovereignty and cross-border data regulations to ensure compliance	3	5	-	2
PC8. handle and document incidents related to non-compliance, establishing appropriate corrective actions	2	4	-	1
Documentation	**5**	**10**	**-**	**5**
PC9. define policies and guidelines on best IT practices that comply with necessary regulatory requirements	1	2	-	1
PC10. document and implement appropriate regulatory and compliance standards that meet the organization’s compliance needs	1	3	-	2
PC11. . create documentation on data sovereignty and cross-border regulations affecting the organization	2	3	-	1

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. develop and maintain a process for tracking compliance issues and mitigation efforts	1	2	-	1
NOS Total	**30**	**50**	**-**	**20**

National Occupational Standards (NOS) Parameters

NOS Code	SSC/N8411
NOS Name	Ensure compliance of regulatory standards, best practices and policies in software development across the organization
Sector	IT-ITeS
Sub-Sector	Future Skills
Occupation	Web & Mobile Development
NSQF Level	6
Credits	2
Version	2.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

SSC/N8452: Optimize Revenue, Security, and Accessibility in Web and Mobile Products

Description

This unit is about defining the key competencies required for optimizing revenue generation, ensuring robust product security, protecting data privacy, and enhancing accessibility in web and mobile products.

Scope

The scope covers the following :

- Revenue Strategy
- Product Solution Strategy
- Customer Requirement Analysis
- Product Security
- Data Privacy and Security
- Accessibility in Web and Mobile Products

Elements and Performance Criteria

Revenue Strategy

To be competent, the user/individual on the job must be able to:

PC1. develop and implement monetization models (e.g., subscriptions, ads, freemium models) based on market analysis and revenue opportunities

PC2. conduct A/B testing and use analytics tools to evaluate product performance and optimize revenue potential

Product Solution Strategy

To be competent, the user/individual on the job must be able to:

PC3. design and prioritize product features aligned with business goals, ROI, and customer needs

PC4. coordinate with cross-functional teams to ensure timely and efficient delivery of product solutions

Customer Requirement Analysis

To be competent, the user/individual on the job must be able to:

PC5. conduct surveys, interviews, and usability testing to gather customer insights and establish continuous feedback loops for iterative improvements

PC6. translate customer requirements into actionable product specifications

Product Security

To be competent, the user/individual on the job must be able to:

PC7. identify and address potential security threats and vulnerabilities through regular audits, penetration testing, and security protocols (e.g., encryption, secure authentication, API security)

Data Privacy and Security

To be competent, the user/individual on the job must be able to:

PC8. adhere to global data privacy regulations (e.g., GDPR, CCPA) by establishing clear policies for data collection, storage, and usage

PC9. ensure compliance with data anonymization and encryption standards

Accessibility in Web and Mobile Products

To be competent, the user/individual on the job must be able to:

PC10. implement and audit accessibility guidelines (e.g., WCAG, ADA) to ensure compliance across web and mobile platforms

PC11. develop inclusive solutions, such as screen reader compatibility, voice commands, and alternative input methods

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. the various monetization models (e.g., subscription-based, freemium, ad-supported) and techniques for analyzing market opportunities and user behavior patterns to optimize revenue

KU2. the product lifecycle management and prioritization frameworks (e.g., MoSCoW, RICE) to guide feature development and iterative improvements

KU3. the methods for collecting and analyzing customer insights (e.g., surveys, interviews) and translating them into actionable product specifications using tools like Jira or Trello

KU4. the common security threats (e.g., phishing, SQL injection) and the application of secure coding practices, vulnerability testing tools, and compliance with standards such as ISO 27001

KU5. the global data privacy laws (e.g., GDPR, CCPA) and principles of data anonymization, encryption, and secure API integration for maintaining transparency and compliance

KU6. the principles of accessibility guidelines (e.g., WCAG, ADA) and inclusive design practices, supported by tools like Lighthouse, to ensure usability for diverse user groups

KU7. the digital marketing strategies and emerging trends in web and mobile technologies (e.g., AI, AR/VR) to align product solutions with market demands

KU8. the integration of analytics tools and dashboards to monitor KPIs related to revenue, security, and accessibility for data-driven decision-making

KU9. the project management methodologies (e.g., Agile, Scrum) and effective communication techniques to coordinate cross-functional teams and stakeholders

KU10. the customer-centric approaches and problem-solving techniques to address complex product challenges and prioritize user needs in development strategies

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. seek clarification and advice from relevant individuals when needed

GS2. listen actively and communicate information clearly and accurately

GS3. adhere to structured decision-making frameworks and guidelines

GS4. make well-informed decisions based on available data and context

GS5. plan tasks effectively to meet targets and deadlines
GS6. apply problem-solving techniques to address various situations
GS7. organize and share relevant information clearly with others
GS8. use balanced judgment to evaluate situations and make decisions

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Revenue Strategy	**5**	**10**	**-**	**4**
PC1. develop and implement monetization models (e.g., subscriptions, ads, freemium models) based on market analysis and revenue opportunities	3	5	-	2
PC2. conduct A/B testing and use analytics tools to evaluate product performance and optimize revenue potential	2	5	-	2
Product Solution Strategy	**5**	**10**	**-**	**4**
PC3. design and prioritize product features aligned with business goals, ROI, and customer needs	3	5	-	2
PC4. coordinate with cross-functional teams to ensure timely and efficient delivery of product solutions	2	5	-	2
Customer Requirement Analysis	**5**	**10**	**-**	**3**
PC5. conduct surveys, interviews, and usability testing to gather customer insights and establish continuous feedback loops for iterative improvements	3	5	-	2
PC6. translate customer requirements into actionable product specifications	2	5	-	1
Product Security	**5**	**5**	**-**	**3**
PC7. identify and address potential security threats and vulnerabilities through regular audits, penetration testing, and security protocols (e.g., encryption, secure authentication, API security)	5	5	-	3
Data Privacy and Security	**5**	**10**	**-**	**3**
PC8. adhere to global data privacy regulations (e.g., GDPR, CCPA) by establishing clear policies for data collection, storage, and usage	3	5	-	2

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC9. ensure compliance with data anonymization and encryption standards	2	5	-	1
Accessibility in Web and Mobile Products	**5**	**5**	**-**	**3**
PC10. implement and audit accessibility guidelines (e.g., WCAG, ADA) to ensure compliance across web and mobile platforms	3	3	-	2
PC11. develop inclusive solutions, such as screen reader compatibility, voice commands, and alternative input methods	2	2	-	1
NOS Total	**30**	**50**	**-**	**20**

National Occupational Standards (NOS) Parameters

NOS Code	SSC/N8452
NOS Name	Optimize Revenue, Security, and Accessibility in Web and Mobile Products
Sector	IT-ITeS
Sub-Sector	
Occupation	Web & Mobile Development
NSQF Level	5
Credits	1
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

DGT/VSQ/N0102: Employability Skills (60 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

PC1. identify employability skills required for jobs in various industries

PC2. identify and explore learning and employability portals

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.

PC4. follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

PC5. recognize the significance of 21st Century Skills for employment

PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

Basic English Skills

To be competent, the user/individual on the job must be able to:

PC7. use basic English for everyday conversation in different contexts, in person and over the telephone

PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English

PC9. write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

PC10. understand the difference between job and career

PC11. prepare a career development plan with short- and long-term goals, based on aptitude

Communication Skills

To be competent, the user/individual on the job must be able to:

PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings

PC13. work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

PC14. communicate and behave appropriately with all genders and PwD

PC15. escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

PC16. select financial institutions, products and services as per requirement

PC17. carry out offline and online financial transactions, safely and securely

PC18. identify common components of salary and compute income, expenses, taxes, investments etc

PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

PC20. operate digital devices and carry out basic internet operations securely and safely

PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively

PC22. use basic features of word processor, spreadsheets, and presentations

Entrepreneurship

To be competent, the user/individual on the job must be able to:

PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research

PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion

PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

PC26. identify different types of customers

PC27. identify and respond to customer requests and needs in a professional manner.

PC28. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC29. create a professional Curriculum vitae (Résumé)

PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC31. apply to identified job openings using offline /online methods as per requirement

PC32. answer questions politely, with clarity and confidence, during recruitment and selection

PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU6. importance of career development and setting long- and short-term goals

KU7. about effective communication

KU8. POSH Act

KU9. Gender sensitivity and inclusivity

KU10. different types of financial institutes, products, and services

KU11. how to compute income and expenditure

KU12. importance of maintaining safety and security in offline and online financial transactions

KU13. different legal rights and laws

KU14. different types of digital devices and the procedure to operate them safely and securely

KU15. how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

KU16. how to identify business opportunities

KU17. types and needs of customers

KU18. how to apply for a job and prepare for an interview

KU19. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence

GS2. communicate effectively using appropriate language in formal and informal settings

GS3. behave politely and appropriately with all
GS4. how to work in a virtual mode
GS5. perform calculations efficiently
GS6. solve problems effectively
GS7. pay attention to details
GS8. manage time efficiently
GS9. maintain hygiene and sanitization to avoid infection

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Introduction to Employability Skills	**1**	**1**	**-**	**-**
PC1. identify employability skills required for jobs in various industries	-	-	-	-
PC2. identify and explore learning and employability portals	-	-	-	-
Constitutional values – Citizenship	**1**	**1**	**-**	**-**
PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC4. follow environmentally sustainable practices	-	-	-	-
Becoming a Professional in the 21st Century	**2**	**4**	**-**	**-**
PC5. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
Basic English Skills	**2**	**3**	**-**	**-**
PC7. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-
PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC9. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
Career Development & Goal Setting	**1**	**2**	**-**	**-**

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. understand the difference between job and career	-	-	-	-
PC11. prepare a career development plan with short- and long-term goals, based on aptitude	-	-	-	-
Communication Skills	**2**	**2**	**-**	**-**
PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings	-	-	-	-
PC13. work collaboratively with others in a team	-	-	-	-
Diversity & Inclusion	**1**	**2**	**-**	**-**
PC14. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC15. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
Financial and Legal Literacy	**2**	**3**	**-**	**-**
PC16. select financial institutions, products and services as per requirement	-	-	-	-
PC17. carry out offline and online financial transactions, safely and securely	-	-	-	-
PC18. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
Essential Digital Skills	**3**	**4**	**-**	**-**
PC20. operate digital devices and carry out basic internet operations securely and safely	-	-	-	-
PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively	-	-	-	-
PC22. use basic features of word processor, spreadsheets, and presentations	-	-	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Entrepreneurship	**2**	**3**	**-**	**-**
PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
Customer Service	**1**	**2**	**-**	**-**
PC26. identify different types of customers	-	-	-	-
PC27. identify and respond to customer requests and needs in a professional manner.	-	-	-	-
PC28. follow appropriate hygiene and grooming standards	-	-	-	-
Getting ready for apprenticeship & Jobs	**2**	**3**	**-**	**-**
PC29. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC31. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC32. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	**20**	**30**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0102
NOS Name	Employability Skills (60 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	12/03/2026
Next Review Date	12/03/2029
NSQC Clearance Date	12/03/2026

SSC/N8408: Define the architecture for the web-based solution

Description

This unit is about defining, designing and maintaining architecture for the web-based solution

Scope

The scope covers the following :

- Shortlisting and Assessment of tools, technologies, frameworks and APIs/ third party service providers
- Scalable and available architecture design
- Risk mitigation
- Maintenance of architecture

Elements and Performance Criteria

Assess tools, technologies, frameworks, and third-party APIs/services

To be competent, the user/individual on the job must be able to:

PC1. Delineate the client-side components, server-side components and other structural components for the web-based solution

PC2. Specify the methodologies for evaluating tools, frameworks and APIs/third party service to be implemented in the product/solution

PC3. Establish important parameters to be analyzed for shortlisting suitable tools, frameworks and APIs/third-party service providers

PC4. Collect and compare data on the tools, frameworks and APIs/third-party service providers in consideration

PC5. Assess long-term dependency risk associated with a tool, framework, API/ service provider to ensure business continuity

PC6. ascertain the security and compliance risks associated with different APIs, tools, frameworks, and third-party services

PC7. assess business continuity for the tools, frameworks, APIs, and service providers, ensuring support and a clear roadmap

PC8. Gauge performance and SLA compliance against business goals for the selected tools, framework, APIs and third-party service providers

Design scalable architecture design

To be competent, the user/individual on the job must be able to:

PC9. Define a suitable technology stack for the web-based solution (such as MEAN, LAMP, MERN etc.)

PC10. Design a suitable architecture for the web applications (such as single-page application, microservices, server-less architecture, monolithic architecture etc.)

PC11. Determine a suitable front-end and back-end web-framework for building the web-solution

PC12. Delineate the services (such database transactions, messaging etc.) and re-usable APIs to be developed as part of the application architecture design

PC13. Define the application boundaries, and its integration aspects to adjacent systems

PC14. Gather the capacity and scalability requirements of the application

PC15. Determine the application's intended operating environment (such as OS types, devices etc.)

PC16. Identify any maintainability, portability and security requirements associated with the application

PC17. Specify the data model for the application/solution

PC18. Design application architecture that is scalable, fault-tolerant, highly-available and resilient

PC19. Decide on suitable load balancing tools to ensure continuous availability of the web-based solution

PC20. Establish application configuration standards that enable runtime configuration changes to applications/solutions

Identify and mitigate risks

To be competent, the user/individual on the job must be able to:

PC21. Specify the Identity and Access Management (IAM) guidelines

PC22. conduct risk assessments and develop a risk management strategy

PC23. Develop a test strategy for the software architecture

Maintain web solution architecture

To be competent, the user/individual on the job must be able to:

PC24. Document and maintain the solution architecture implemented in the organization

PC25. Review existing system architecture designs to ensure balance of service quality, use of appropriate technology and efficient utilization of resources

PC26. Monitor changes in solution design standards and update architecture accordingly

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Organizational policies, procedures and guidelines which relate to defining requirements for the solution

KU2. Organizational policies and procedures for shortlisting third-party service providers

KU3. Organizational policies and procedures for documenting various standards, regulations and guidelines

KU4. Whom to involve while maintaining and reviewing system architecture

KU5. The range of standard templates and tools available and how to use them

KU6. Best practices for designing, developing and maintaining software solutions

KU7. Different technology stacks for developing solutions

KU8. Different front-end and back-end web-frameworks for developing solutions

KU9. Different services that can be used as part of architecture design

KU10. Different operating environments for developing solutions

KU11. how to design cloud-native applications, focusing on microservices architecture and its benefits in scalability and maintenance

KU12. the disaster recovery lifecycle and its importance in application resilience

KU13. the use of tools such as Selenium, JMeter, and JUnit to ensure the reliability of web-based solutions by automating unit testing, integration testing, and performance testing

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. follow guidelines, procedures, and rules to ensure compliance and efficiency
GS2. contribute to team performance by collaborating and sharing knowledge
GS3. work flexibly both independently and with teams
GS4. produce accurate, detailed, and well-written work
GS5. evaluate business impact and share relevant information with stakeholders
GS6. communicate important information clearly and effectively
GS7. ensure work is complete and error-free before submission
GS8. apply critical thinking to make balanced decisions
GS9. provide detailed, constructive feedback and solutions
GS10. make informed decisions considering both short- and long-term goals
GS11. plan and organize work to meet targets and deadlines
GS12. apply problem-solving strategies to resolve issues effectively

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Assess tools, technologies, frameworks, and third-party APIs/services	**10**	**15**	**-**	**8**
PC1. Delineate the client-side components, server-side components and other structural components for the web-based solution	1	1	-	1
PC2. Specify the methodologies for evaluating tools, frameworks and APIs/third party service to be implemented in the product/solution	2	2	-	1
PC3. Establish important parameters to be analyzed for shortlisting suitable tools, frameworks and APIs/third-party service providers	1	2	-	1
PC4. Collect and compare data on the tools, frameworks and APIs/third-party service providers in consideration	1	2	-	1
PC5. Assess long-term dependency risk associated with a tool, framework, API/ service provider to ensure business continuity	2	2	-	1
PC6. ascertain the security and compliance risks associated with different APIs, tools, frameworks, and third-party services	1	2	-	1
PC7. assess business continuity for the tools, frameworks, APIs, and service providers, ensuring support and a clear roadmap	1	2	-	1
PC8. Gauge performance and SLA compliance against business goals for the selected tools, framework, APIs and third-party service providers	1	2	-	1
Design scalable architecture design	**14**	**20**	**-**	**10**
PC9. Define a suitable technology stack for the web-based solution (such as MEAN, LAMP, MERN etc.)	1	1	-	1
PC10. Design a suitable architecture for the web applications (such as single-page application, microservices, server-less architecture, monolithic architecture etc.)	1	2	-	1

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. Determine a suitable front-end and back-end web-framework for building the web-solution	2	1	-	1
PC12. Delineate the services (such database transactions, messaging etc.) and re-usable APIs to be developed as part of the application architecture design	1	2	-	1
PC13. Define the application boundaries, and its integration aspects to adjacent systems	1	2	-	1
PC14. Gather the capacity and scalability requirements of the application	1	2	-	-
PC15. Determine the application's intended operating environment (such as OS types, devices etc.)	1	2	-	1
PC16. Identify any maintainability, portability and security requirements associated with the application	1	2	-	1
PC17. Specify the data model for the application/solution	2	1	-	1
PC18. Design application architecture that is scalable, fault-tolerant, highly-available and resilient	1	2	-	1
PC19. Decide on suitable load balancing tools to ensure continuous availability of the web-based solution	1	1	-	1
PC20. Establish application configuration standards that enable runtime configuration changes to applications/solutions	1	2	-	-
Identify and mitigate risks	**3**	**6**	**-**	**1**
PC21. Specify the Identity and Access Management (IAM) guidelines	1	2	-	-
PC22. conduct risk assessments and develop a risk management strategy	1	2	-	-
PC23. Develop a test strategy for the software architecture	1	2	-	1

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Maintain web solution architecture	**3**	**9**	**-**	**1**
PC24. Document and maintain the solution architecture implemented in the organization	1	3	-	-
PC25. Review existing system architecture designs to ensure balance of service quality, use of appropriate technology and efficient utilization of resources	1	3	-	1
PC26. Monitor changes in solution design standards and update architecture accordingly	1	3	-	-
NOS Total	**30**	**50**	**-**	**20**

National Occupational Standards (NOS) Parameters

NOS Code	SSC/N8408
NOS Name	Define the architecture for the web-based solution
Sector	IT-ITeS
Sub-Sector	Future Skills
Occupation	Web & Mobile Development
NSQF Level	6
Credits	3
Version	2.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

SSC/N8412: Define security architecture and security controls to ensure security of the web-based solution

Description

This unit is about defining security architecture and controls for the web-based solution.

Scope

The scope covers the following :

- Define security architecture and controls

Elements and Performance Criteria

Define security architecture and controls

To be competent, the user/individual on the job must be able to:

PC1. Determine the security requirements of the customer/organisation and internal departments

PC2. Design the security architecture for the web-based solution

PC3. Research, identify and implement security protocols (such as SSL, TLS, PGP, IPsec etc.) And other best practices to ensure the security of the web-based product

PC4. Identify and adopt security standards (such as OWASP and other applicable standards.), policies and protocols appropriate for the solution

PC5. Conduct security tests (such as threat analysis, vulnerability scanning, penetration testing, risk assessment etc.) to identify security vulnerabilities

PC6. Establish security safeguards to be implemented (such as access, data, architecture, infrastructure etc.)

PC7. Specify the Identity and Access Management (IAM) guidelines

PC8. Elucidate the policies for data leakage prevention (DLP)

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Organisational policies, procedures and guidelines which relate to internal security standards, regulations and guidelines

KU2. Organisational policies and procedures which relate to implementing security architecture

KU3. Organisational policies and procedures for documenting various security standards, regulations and guidelines

KU4. Whom to involve while conducting security tests

KU5. The range of standard templates and tools available and how to use them

KU6. Best practices for ensuring security of the solution

KU7. Different security protocols for web-based solutions

KU8. How to conduct security tests

KU9. How to define security guidelines and policies

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Follow instructions, guidelines, procedures, rules and service level agreements
GS2. Contribute to the quality of team-work
GS3. Work independently and collaboratively
GS4. Complete accurate well written work with attention to detail
GS5. Understand business impact and disseminate relevant information to others
GS6. Pass on relevant information to others
GS7. Check the work is complete and free from errors
GS8. Apply balanced judgments to different situations
GS9. Provide opinions on work in a detailed and constructive way
GS10. Make decisions on suitable courses
GS11. Plan and organize the work to achieve targets and deadlines
GS12. Apply problem-solving approaches in different situations

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Define security architecture and controls	**30**	**50**	**-**	**20**
PC1. Determine the security requirements of the customer/organisation and internal departments	3	5	-	2
PC2. Design the security architecture for the web-based solution	4	6	-	2
PC3. Research, identify and implement security protocols (such as SSL, TLS, PGP, IPsec etc.) And other best practices to ensure the security of the web-based product	5	7	-	3
PC4. Identify and adopt security standards (such as OWASP and other applicable standards.), policies and protocols appropriate for the solution	5	7	-	3
PC5. Conduct security tests (such as threat analysis, vulnerability scanning, penetration testing, risk assessment etc.) to identify security vulnerabilities	4	7	-	3
PC6. Establish security safeguards to be implemented (such as access, data, architecture, infrastructure etc.)	3	7	-	3
PC7. Specify the Identity and Access Management (IAM) guidelines	3	6	-	2
PC8. Elucidate the policies for data leakage prevention (DLP)	3	5	-	2
NOS Total	**30**	**50**	**-**	**20**

National Occupational Standards (NOS) Parameters

NOS Code	SSC/N8412
NOS Name	Define security architecture and security controls to ensure security of the web-based solution
Sector	IT-ITeS
Sub-Sector	Future Skills
Occupation	Web & Mobile Development
NSQF Level	6
Credits	2
Version	2.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

SSC/N8409: Define the architecture for the mobile-based solution

Description

This unit is about defining, designing and maintaining architecture for the mobile-based solution

Scope

The scope covers the following :

- Shortlisting and Assessment of tools, technologies, frameworks and APIs/ third party service providers
- Architecture design
- Risk mitigation
- Maintenance of architecture

Elements and Performance Criteria

Assess tools, technologies, frameworks, and third-party APIs/services

To be competent, the user/individual on the job must be able to:

PC1. Specify the methodologies for evaluating tools, frameworks and APIs/third party service to be implemented in the product/solution

PC2. Establish important parameters to be analyzed for shortlisting suitable tools, frameworks and APIs/third-party service providers

PC3. Collect and compare data on the tools, frameworks and APIs/third-party service providers in consideration

PC4. Assess long-term dependency risk associated with a tool, framework, API/ service provider to ensure business continuity

PC5. Understand the security and compliance risks associated with different APIs/tools/frameworks/third party services

PC6. assess business continuity aspects, ensuring support and roadmap alignment with tools, frameworks, APIs, or service providers

PC7. gauge performance and SLA compliance of tools, frameworks, APIs, and third-party service providers based on business goals.

Design architecture for mobile-based solutions

To be competent, the user/individual on the job must be able to:

PC8. Specify the target platforms (such as Android, iOS, SailfishOS, KaiOS etc.) for the mobile-based solution

PC9. define the type of mobile application (native, hybrid, cross-platform) according to functional and non-functional requirements.

PC10. Delineate different client-side components, server-side components and other structural components for the mobile solution

PC11. Identify different types of system level/hardware level access provided by the target mobile platform

PC12. design a suitable technology stack for the mobile-based solution based on requirements

PC13. Prescribe a suitable framework for developing application for the target mobile platform

PC14. design services (e.g., database transactions, messaging) and re-usable APIs as part of application architecture

PC15. define application boundaries and integration points with adjacent systems

PC16. gather capacity and scalability requirements for the mobile application.

PC17. Determine the application's intended operating environment (such as OS types, devices etc.)

PC18. Identify any maintainability, portability and security requirements associated with the application

PC19. design the data model for the mobile application/solution

PC20. Establish the application configuration standards that enable runtime configuration changes to applications/solutions

Design user-centric UX/UI interfaces

To be competent, the user/individual on the job must be able to:

PC21. apply foundational concepts of UX/UI design to ensure a user-friendly and intuitive interface

PC22. design interfaces based on user needs, personas, and customer journeys, ensuring accessibility and responsiveness

PC23. integrate feedback from user testing and iterative design to refine and enhance the user experience

Optimize mobile application performance

To be competent, the user/individual on the job must be able to:

PC24. implement performance optimization methods such as caching, efficient data management, and lightweight asset usage

PC25. conduct stress and load testing to assess performance under varying conditions

PC26. use mobile-specific performance monitoring tools to identify bottlenecks and implement performance enhancements.

Apply agile methodologies for mobile development

To be competent, the user/individual on the job must be able to:

PC27. apply Agile methodologies, focusing on iterative development cycles, user feedback, and continuous improvement

PC28. break down tasks into sprints and manage backlogs, ensuring timely delivery and adaptability to user feedback.

PC29. collaborate with cross-functional teams (development, design, QA) to enhance speed and efficiency.

Identify and mitigate risks

To be competent, the user/individual on the job must be able to:

PC30. develop a comprehensive test strategy for mobile application architecture

PC31. align architecture with regulatory standards and compliance requirements.

PC32. identify application dependencies and conduct thorough risk assessments of mobile solutions.

Maintain mobile solution architecture

To be competent, the user/individual on the job must be able to:

PC33. document and maintain the solution architecture implemented in the organization

PC34. review existing system architecture designs to ensure a balance between service quality, technology appropriateness, and resource efficiency.

PC35. monitor and implement changes in solution design standards to keep the architecture updated.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. the procedures for defining requirements for mobile solutions
KU2. the procedures for shortlisting third-party service providers
KU3. the procedures for documenting standards, regulations, and guidelines
KU4. the roles involved in maintaining and reviewing system architecture
KU5. the standard templates and tools available, and how to use them effectively
KU6. the best practices for designing, developing, and maintaining software solutions
KU7. different technology stacks for developing mobile applications
KU8. front-end and back-end web frameworks for mobile solutions
KU9. services that can be used as part of the mobile architecture design
KU10. services that can be used as part of the mobile architecture design
KU11. the foundational UX/UI design principles and how to apply them to mobile solutions
KU12. performance optimization methods for mobile apps
KU13. agile methodologies and how to apply them in mobile application development

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. follow instructions, guidelines, and procedures, adhering to rules and service level agreements
GS2. contribute effectively to team performance
GS3. work well independently and with others
GS4. produce accurate, detailed, and well-written work
GS5. assess business impact and share relevant information
GS6. communicate important information clearly
GS7. ensure work is complete and error-free
GS8. make fair and balanced judgments
GS9. provide constructive and detailed feedback
GS10. make informed decisions on the best course of action
GS11. plan and organize tasks to meet deadlines
GS12. apply problem-solving skills effectively

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Assess tools, technologies, frameworks, and third-party APIs/services	**7**	**12**	**-**	**4**
PC1. Specify the methodologies for evaluating tools, frameworks and APIs/third party service to be implemented in the product/solution	1	1	-	0.5
PC2. Establish important parameters to be analyzed for shortlisting suitable tools, frameworks and APIs/third-party service providers	1	2	-	0.5
PC3. Collect and compare data on the tools, frameworks and APIs/third-party service providers in consideration	1	2	-	1
PC4. Assess long-term dependency risk associated with a tool, framework, API/ service provider to ensure business continuity	1	2	-	0.5
PC5. Understand the security and compliance risks associated with different APIs/tools/frameworks/third party services	1	2	-	0.5
PC6. assess business continuity aspects, ensuring support and roadmap alignment with tools, frameworks, APIs, or service providers	1	2	-	0.5
PC7. gauge performance and SLA compliance of tools, frameworks, APIs, and third-party service providers based on business goals.	1	1	-	0.5
Design architecture for mobile-based solutions	**11**	**21**	**-**	**10**
PC8. Specify the target platforms (such as Android, iOS, SailfishOS, KaiOS etc.) for the mobile-based solution	1	2	-	1
PC9. define the type of mobile application (native, hybrid, cross-platform) according to functional and non-functional requirements.	1	2	-	1
PC10. Delineate different client-side components, server-side components and other structural components for the mobile solution	1	2	-	1

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. Identify different types of system level/hardware level access provided by the target mobile platform	1	2	-	0.5
PC12. design a suitable technology stack for the mobile-based solution based on requirements	1	2	-	1
PC13. Prescribe a suitable framework for developing application for the target mobile platform	1	2	-	1
PC14. design services (e.g., database transactions, messaging) and re-usable APIs as part of application architecture	1	2	-	1
PC15. define application boundaries and integration points with adjacent systems	0.5	2	-	1
PC16. gather capacity and scalability requirements for the mobile application.	0.5	1	-	0.5
PC17. Determine the application's intended operating environment (such as OS types, devices etc.)	1	1	-	0.5
PC18. Identify any maintainability, portability and security requirements associated with the application	0.5	1	-	0.5
PC19. design the data model for the mobile application/solution	1	1	-	0.5
PC20. Establish the application configuration standards that enable runtime configuration changes to applications/solutions	0.5	1	-	0.5
Design user-centric UX/UI interfaces	**3**	**4**	**-**	**2**
PC21. apply foundational concepts of UX/UI design to ensure a user-friendly and intuitive interface	1	2	-	1
PC22. design interfaces based on user needs, personas, and customer journeys, ensuring accessibility and responsiveness	1	1	-	0.5
PC23. integrate feedback from user testing and iterative design to refine and enhance the user experience	1	1	-	0.5

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Optimize mobile application performance	**3**	**4**	**-**	**1**
PC24. implement performance optimization methods such as caching, efficient data management, and lightweight asset usage	1	2	-	-
PC25. conduct stress and load testing to assess performance under varying conditions	1	1	-	0.5
PC26. use mobile-specific performance monitoring tools to identify bottlenecks and implement performance enhancements.	1	1	-	0.5
Apply agile methodologies for mobile development	**2**	**3**	**-**	**1**
PC27. apply Agile methodologies, focusing on iterative development cycles, user feedback, and continuous improvement	1	1	-	0.5
PC28. break down tasks into sprints and manage backlogs, ensuring timely delivery and adaptability to user feedback.	0.5	1	-	0.5
PC29. collaborate with cross-functional teams (development, design, QA) to enhance speed and efficiency.	0.5	1	-	-
Identify and mitigate risks	**2**	**3**	**-**	**1**
PC30. develop a comprehensive test strategy for mobile application architecture	1	1	-	0.5
PC31. align architecture with regulatory standards and compliance requirements.	0.5	1	-	-
PC32. identify application dependencies and conduct thorough risk assessments of mobile solutions.	0.5	1	-	0.5
Maintain mobile solution architecture	**2**	**3**	**-**	**1**
PC33. document and maintain the solution architecture implemented in the organization	0.5	1	-	0.5
PC34. review existing system architecture designs to ensure a balance between service quality, technology appropriateness, and resource efficiency.	0.5	1	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC35. monitor and implement changes in solution design standards to keep the architecture updated.	1	1	-	0.5
NOS Total	**30**	**50**	**-**	**20**

National Occupational Standards (NOS) Parameters

NOS Code	SSC/N8409
NOS Name	Define the architecture for the mobile-based solution
Sector	IT-ITeS
Sub-Sector	Future Skills
Occupation	Web & Mobile Development
NSQF Level	6
Credits	2
Version	3.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

SSC/N8413: Define security architecture and security controls to ensure security of the mobile application/solution

Description

This unit is about defining security architecture and controls for the mobile-based solution

Scope

The scope covers the following :

- Define security architecture and controls

Elements and Performance Criteria

Define security architecture and controls

To be competent, the user/individual on the job must be able to:

PC1. Determine the security requirements of the customer/organisation and internal departments
PC2. Specify the security provisions of the target mobile platforms
PC3. Test application code for security vulnerabilities
PC4. Secure the application code through appropriate encryptions
PC5. Establish identification, authentication and authorization measures
PC6. Ensure third-party APIs are secure and comply by the organizations security standards
PC7. Identify and adopt security standards (such as OWASP, CVSS, CWE etc.), policies and protocols appropriate for the solution
PC8. Implement appropriate security controls to secure the data stored in the device
PC9. Specify Identity and Access Management (IAM) guidelines
PC10. Elucidate policies for data leakage prevention (DLP)

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Organisational policies, procedures and guidelines which relate to internal security standards, regulations and guidelines
KU2. Organisational policies and procedures which relate to implementing security architecture
KU3. Organisational policies and procedures for documenting various security standards, regulations and guidelines
KU4. Whom to involve while conducting security tests
KU5. The range of standard templates and tools available and how to use them
KU6. Best practices for ensuring security of the solution
KU7. different security protocols for mobile-based solutions
KU8. How to conduct security tests

KU9. How to define security guidelines and policies

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Follow instructions, guidelines, procedures, rules and service level agreements
GS2. Contribute to the quality of team-work
GS3. Work independently and collaboratively
GS4. Complete accurate well written work with attention to detail
GS5. Understand business impact and disseminate relevant information to others
GS6. Pass on relevant information to others
GS7. Check the work is complete and free from errors
GS8. Apply balanced judgments to different situations
GS9. Provide opinions on work in a detailed and constructive way
GS10. Make decisions on suitable courses
GS11. Plan and organize the work to achieve targets and deadlines
GS12. Apply problem-solving approaches in different situations

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Define security architecture and controls	**30**	**50**	**-**	**20**
PC1. Determine the security requirements of the customer/organisation and internal departments	2	4	-	2
PC2. Specify the security provisions of the target mobile platforms	3	4	-	2
PC3. Test application code for security vulnerabilities	4	7	-	2
PC4. Secure the application code through appropriate encryptions	3	5	-	2
PC5. Establish identification, authentication and authorization measures	3	5	-	2
PC6. Ensure third-party APIs are secure and comply by the organizations security standards	3	7	-	2
PC7. Identify and adopt security standards (such as OWASP, CVSS, CWE etc.), policies and protocols appropriate for the solution	4	7	-	2
PC8. Implement appropriate security controls to secure the data stored in the device	2	4	-	2
PC9. Specify Identity and Access Management (IAM) guidelines	3	4	-	2
PC10. Elucidate policies for data leakage prevention (DLP)	3	3	-	2
NOS Total	**30**	**50**	**-**	**20**

National Occupational Standards (NOS) Parameters

NOS Code	SSC/N8413
NOS Name	Define security architecture and security controls to ensure security of the mobile application/solution
Sector	IT-ITeS
Sub-Sector	Future Skills
Occupation	Web & Mobile Development
NSQF Level	6
Credits	2
Version	2.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

The assessment shall be conducted through an online proctored format, incorporating scenario-based multiple-choice questions designed to effectively evaluate practical understanding and real-world application of concepts. Additionally, it will include a viva-voice and hands-on practical evaluation to comprehensively assess the individual's proficiency in specific learning outcomes.

Minimum Aggregate Passing % at QP Level : 70

(**Please note**: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
SSC/N8407.Define best practices, checklist, reference implementations for application design and development	30	50	-	20	100	20
SSC/N8306.Translate business-level use cases into technical requirements and specifications	30	50	-	20	100	20
SSC/N8410.Define parameters and tools to monitor and manage application performance	30	50	-	20	100	20
SSC/N8411.Ensure compliance of regulatory standards, best practices and policies in software development across the organization	30	50	-	20	100	20
SSC/N8452.Optimize Revenue, Security, and Accessibility in Web and Mobile Products	30	50	-	20	100	5
DGT/VSQ/N0102.Employability Skills (60 Hours)	20	30	-	-	50	5
Total	**170**	**280**	**-**	**100**	**550**	**90**

Elective: 1 Web Application Architecture

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
SSC/N8408.Define the architecture for the web-based solution	30	50	-	20	100	5
SSC/N8412.Define security architecture and security controls to ensure security of the web-based solution	30	50	-	20	100	5
Total	**60**	**100**	**-**	**40**	**200**	**10**

Elective: 2 Mobile Application Architecture

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
SSC/N8409.Define the architecture for the mobile-based solution	30	50	-	20	100	5
SSC/N8413.Define security architecture and security controls to ensure security of the mobile application/solution	30	50	-	20	100	5
Total	**60**	**100**	**-**	**40**	**200**	**10**

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training
NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
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NOS	National Occupational Standard(s)
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NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.
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Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.
Knowledge and understanding	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual need in order to perform to the required standard.
Organisational context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core skills/ Generic skills	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialised job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.
Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.

Occupational standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standard	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications pack (QP)	QP comprises the set of OSs, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
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